

1.17 Solution: (a) The electrostatic energy is given by

$$W = \frac{\epsilon_0}{2} \int_V |\nabla \Phi|^2 d^3x.$$

By the definition of capacitance, Eq. (1.62),

$$W = \frac{1}{2} \sum_{ij} C_{ij} V_i V_j = \frac{C}{2},$$

since we have only one conductor with $V = 1$ and others with $V = 0$. Therefore,

$$C = \epsilon_0 \int_V |\nabla \Phi|^2 d^3x.$$

(b) For any trial function Ψ , we have

$$W[\Psi] = \frac{\epsilon_0}{2} \int_V |\nabla \Psi|^2 d^3x \geq W.$$

Then, we must have

$$C[\Psi] = \epsilon_0 \int_V |\nabla \Phi|^2 d^3x \geq C.$$